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RESEARCH PAPER

Comparative thermal and computational analysis of the WAAM process using FEM and FDM models with experimental validation

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Abstract

This study investigates the thermal behavior of a WAAM-deposited single bead through two different numerical modelling strategies: Finite Element Method (FEM) and Finite Difference Method (FDM). There is still a lack of direct comparative studies between different discretization strategies applied under identical experimental conditions. In particular, the trade-off between prediction accuracy and computational efficiency in FEM and FDM thermal models has not been sufficiently quantified for WAAM applications. Both approaches integrate a double double-ellipsoid heat source model to better capture the melt pool geometry, resulting in high agreement with experimental profiles and fusion boundaries. The parameters of the double double-ellipsoid heat source were defined from melt pool boundaries extracted from experimental bead cross-sections using an image-processing procedure. In addition, a comparative analysis of modelling accuracy and computational cost is carried out, showing that while both methods yield reliable predictions and comparable accuracy in the estimation of the melt pool geometry and the heat-affected zone (HAZ), the FDM approach reduces the simulation time by approximately a factor of 30 compared with the FEM model, without compromising prediction quality. These findings highlight the potential of simplified thermal models for industrial implementation, offering a practical balance between precision and efficiency in WAAM process simulation.

Keywords WAAM · FEM · FDM · Thermal modelling · Melt pool

1 Introduction

Additive manufacturing (AM) has emerged as a transformative technology across various industries, enabling the fabrication of complex geometries, reduced material waste, and accelerated production times. Among the different additive manufacturing technologies currently available, processes such as Powder Bed Fusion (PBF), Directed Energy Deposition (DED), Binder Jetting (BJ), and Material Extrusion (ME) have been widely adopted depending on the material system, component size, and application requirements [1–3]. Within this context, DED-Arc (WAAM) stands out due to its high deposition rates, cost-efficiency, and ability to process a wide range of metallic alloys. WAAM is increasingly employed in aerospace, automotive, and marine sectors for manufacturing large-scale structural components and for the repair of high-value parts [4, 5].

However, the process involves complex thermal phenomena, including repetitive heating and cooling cycles, solidification, and the formation of residual stresses, all of which influence the final part quality [6]. Accurate thermal modelling is therefore essential for understanding and controlling these effects. Thermal simulations enable the prediction of temperature fields, cooling rates, and melt pool dimensions, providing valuable insights for process optimization and defect prevention [7, 8]. Several modelling approaches are available to simulate heat transfer in WAAM, such as Finite Element Methods (FEM), Finite Difference Methods

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(FDM), and analytical solutions [9–11]. Each method offers different levels of accuracy, computational cost, and implementation complexity, depending on the specific requirements of the simulation.

Among the different thermal modelling approaches used in WAAM, the Finite Element Method is one of the most widely adopted due to its flexibility and accuracy in handling complex geometries, boundary conditions, and material heterogeneities [12]. Unlike the Finite Difference Method, which is typically limited to structured grids and regular domains, FEM allows for unstructured meshing and greater adaptability to the intricate shapes involved in additive manufacturing. Furthermore, while analytical models offer fast solutions with minimal computational cost, they often rely on strong simplifications that limit their applicability to real-world WAAM scenarios [10, 13]. FEM, in contrast, enables the detailed prediction of temperature fields and melt pool dimensions with higher fidelity, making it particularly suitable for process optimization and validation against experimental data [14]. Mahmood et al. [10] proposed both analytical and FEM-based thermomechanical models to predict thermal fields and residual stresses during multi-pass arc welding with applicability to arc-based additive manufacturing.

The Finite Difference Method is a numerical approach that is employed in thermal simulations of WAAM due to its conceptual simplicity and relatively low computational cost [11]. It discretizes the heat conduction equations over structured grids, making it particularly efficient for problems with simple geometries and uniform mesh structures [15, 16]. Compared to FEM, FDM is generally faster to implement and computationally lighter; however, it is less flexible when dealing with complex geometries, irregular boundaries, or variable material properties. In contrast to analytical models, FDM offers greater accuracy and the ability to incorporate transient effects, but still requires careful meshing and stability considerations [17]. In this context, the work of Stockman et al. [11] presents a valuable reference point for the development of computationally efficient thermal models. Their study introduces a simplified transient thermal model for metal additive manufacturing, leveraging FDM along with coarse spatial and temporal meshing to drastically reduce computational cost. This demonstrates the potential of reduced-complexity models to support process optimization in industrial settings, and motivates the present work. Thermal models such as those developed in the present study have significant industrial potential in WAAM applications, as they enable the prediction of melt pool geometry, heat-affected zone dimensions, and thermal histories under different process conditions.

An accurate representation of the heat source is critical in thermal simulations of WAAM, as it directly influences the predicted melt pool geometry and temperature distribution. Several heat flux source models have been developed to approximate the energy input from the arc, each with varying levels of complexity and applicability depending on the welding process and simulation objectives [18]. Kik et al. [19] highlight that conventional thermal models used for arc-welding processes often fail to accurately capture the geometry of the melt pool. To address this limitation, they proposed three modified configurations of the widely used double-ellipsoidal heat source model originally introduced by Goldak et al. [20]. Among these, one of the models incorporates a second ellipsoidal heat source to better reproduce the melt pool shape, particularly its depth or "finger"-like penetration. Azar et al. [21] developed a two-dimensional thermal model incorporating two perpendicular, split ellipsoidal heat sources, once again based on the Goldak double-ellipsoidal formulation, to better replicate the melt pool penetration, or "finger" morphology, in Gas Metal Arc Welding (GMAW) processes. The parameters defining the heat source geometry were initially obtained from experimental observations, such as the top-view geometry of the terminal crater and subsequently refined through iterative calibration.

Computational cost is a critical factor in selecting an appropriate thermal modelling approach for WAAM processes, as it directly impacts the feasibility of simulations, especially for large-scale or multi-layer builds [11]. FEM, while offering high accuracy and flexibility, generally demands significant computational resources and longer processing times due to complex meshing and the need to solve large systems of equations [22]. In contrast, FDM typically requires less computational effort, benefiting from simpler grid structures and more straightforward implementations, which results in faster runtimes [23]. However, this reduction in computational cost with FDM often comes at the expense of decreased geometric flexibility and lower accuracy in complex domains. Balancing computational efficiency and model fidelity is therefore essential for practical simulation and process optimization in WAAM. Ding et al. [24] highlighted the computational inefficiency of conventional transient thermomechanical FEM models when applied to large-scale WAAM processes. To address this, they developed a simplified FE model that reduced computation time by 99% while maintaining high accuracy in predicting residual stresses and distortions.

In industrial environments, these models can be employed for process parameter optimization, reduction of trial-and-error experimentation, minimization of thermal defects and distortions, and the future implementation of real-time monitoring and control strategies aimed at improving manufacturing efficiency and component quality.

Despite the extensive use of thermal modelling in WAAM, most previous studies have primarily focused on FEM-based approaches, which often involve high computational costs that limit their applicability in process

optimization and real-time industrial implementation. In contrast, simplified approaches such as the Finite Difference Method (FDM) have received significantly less attention, particularly in terms of direct comparison with FEM models under identical experimental conditions. Therefore, there remains a research gap regarding the trade-off between computational efficiency and prediction accuracy when both modelling strategies are applied to WAAM thermal simulations. In this study, an experimental WAAM single bead was produced using specific process parameters, serving as a basis for thermal modelling via two distinct numerical approaches: the Finite Element Method and the Finite Difference Method. The thermal model adopted in the present work closely resembles the explained configuration, which employs a combination of a standard (upper) ellipsoidal heat source and an additional vertically oriented ellipsoid to capture the melt pool's penetration characteristics. The main objective of this work is to compare the thermal prediction capability and computational performance of FEM and FDM approaches through both qualitative and quantitative analyses against experimental results. The novelty of the present research lies in the systematic comparison of both methods using the same experimental conditions and heat source definition, allowing a direct evaluation of their respective advantages and limitations. The results demonstrate that the FDM approach can substantially reduce computational cost while maintaining prediction accuracy comparable to FEM in the estimation of melt pool geometry and heat-affected zone dimensions. These findings provide valuable insight into the development of efficient thermal simulation tools for industrial WAAM process optimization, monitoring, and control.

2 Materials and methods

This section describes the experimental procedure and numerical methodologies employed to analyze the thermal behavior of the WAAM process. First, the deposition of a single bead under controlled parameters is detailed, serving as the reference for numerical validation. Subsequently, the two thermal models developed (based on FEM and FDM) are presented, including the heat source formulation adopted to reproduce melt pool morphology. Finally, the computational analysis strategy is outlined, focusing on various parameters to carry out an evaluation of computational cost.

2.1 Experimental procedure

The additive manufacturing process was carried out using an Addilan WAAM system (Durango, Spain), equipped with a two-axis tilting table and operating inside a sealed chamber under an inert gas environment (Fig. 1). The deposition was performed using Gas Metal Arc Welding (GMAW) as the primary welding method, although the equipment also supports alternative arc-based processes such as Gas Tungsten Arc Welding (GTAW) and Plasma Arc Welding (PAW). The system configuration included a Titan XQ 400 AC Puls MIG/MAG power source and an M drive 4 Rob5 XR RE wire feeder, both supplied by EWM (Mündersbach, Germany). For in-situ monitoring of bead geometry, a compact pyrometer from Optris (Berlin, Germany) and a Q4 laser scanner from QuellTech (Munich, Germany) were mounted directly on the welding torch assembly.



(a) (b)

Fig. 1 **a** General view of the WAAM system used for bead deposition (Addilan equipment). **b** Detail of the deposition zone showing the wire feeding system, nozzle, and bead formation during the WAAM process inside the machine

The substrate used in the experiments measured 120 mm in length, 60 mm in width, and 30 mm in thickness. The WAAM deposition was carried out over an 80 mm path along the longitudinal (lengthwise) direction, centered on the top surface of the substrate to ensure consistent heat distribution and boundary conditions, using a metal wire of 1.2 mm in diameter as the feedstock material. The WAAM deposition was carried out with a deposition rate of 8 kg/h and a travel speed of 150 cm/min. The torch was equipped with a 20 mm diameter nozzle and a stick-out length of 15 mm. Shielding was provided by an inert gas mixture composed of 80% argon and 20% carbon dioxide, delivered at a flow rate of 17 l/min. The welding power applied during the process was 12,757 W, ensuring a stable arc and sufficient energy input for consistent material deposition. The process parameters used in this study were selected based on previous experimental and numerical investigations by the authors, where an optimal set of conditions ensuring stable bead formation and consistent thermal behavior in WAAM was identified [18].

The substrate material used for the WAAM deposition was S235JR structural steel, a commonly employed low-carbon steel due to its good weldability, mechanical strength, and widespread availability. In additive manufacturing applications, particularly WAAM, S235JR serves as a robust base for multi-pass deposition thanks to its thermal and mechanical stability under repeated heating cycles [25]. Its melting temperature is approximately 1450 °C, while the

austenitization or phase transformation range lies between 850 and 900 °C (critical thresholds that were used in the thermal simulations to delineate the melt pool boundaries and the heat-affected zone (HAZ)). As filler material, ER70S-6 welding wire was employed. This solid wire is widely used in WAAM for its excellent deposition characteristics, high ductility, and good strength-to-cost ratio. Its relatively high silicon and manganese content enhances deoxidation and weld pool fluidity, making it particularly suitable for stable and high-quality arc-based additive manufacturing processes [26].

2.2 Numerical modelling

Both the FEM and FDM thermal models consist of a substrate onto which a WAAM single bead is deposited. For the numerical modeling, the substrate dimensions were set to match those of the substrate employed in the experimental work. These thermal models were developed using a consistent set of initial and boundary conditions to ensure comparability and alignment with the experimental setup. The ambient temperature was fixed at 25 °C, corresponding to standard laboratory conditions, and the initial temperature of the substrate was uniformly set to 25 °C, assuming thermal equilibrium prior to the heat input. Convective heat transfer to the surroundings was included as the primary mechanism for energy dissipation, taking into account the presence of an argon-rich atmosphere in the deposition chamber, which reduces convective losses compared to air. In addition to convection, radiative heat losses were also considered in both models, as radiation becomes a non-negligible mechanism of energy dissipation at the high temperatures typical of arc-based processes [27, 28].

With respect to process efficiency, a value of 0.85 was used to define the effective heat input in both models. This value lies at the upper end of the typical efficiency range reported for WAAM processes (0.70–0.85), and reflects not only the energy losses due to radiation and convection (already accounted for separately in the model) but also practical factors such as arc stability, metal transfer mode, and the high deposition efficiency associated with the process parameters employed (short stick-out, high current, and stable arc conditions). These considerations support the adoption of a relatively high efficiency value for representing the actual thermal energy absorbed by the workpiece during deposition [29].

In both the FEM and FDM simulations, material properties such as thermal conductivity, specific heat capacity and density were assumed to be temperature-dependent. These temperature-dependent values were incorporated into the models to more accurately capture the transient thermal behavior during the WAAM process [14].

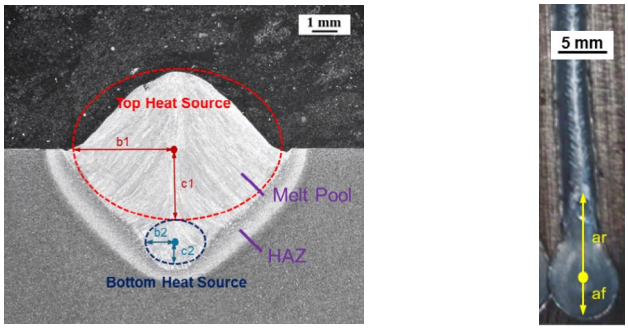


Fig. 2 Experimentally deposited WAAM single bead on the substrate. **a** Cross-sectional view along with both ellipsoid widths and heights. **b** Top view of the final crater along with front and rear length parameters

2.2.1 Heat source model

To accurately replicate the geometry of the melt pool in the thermal simulations, a heat source model capable of capturing both the lateral spread and the depth of the molten region was required. For this purpose, the present study adopts the approach proposed by Kik et al. [19], which employs a double double-ellipsoidal heat source based on the classical model developed by Goldak et al. [20]. In this configuration, the standard horizontal double-ellipsoidal heat source primarily represents the heat distribution near the surface, accounting for the majority of the melt pool volume (Top Heat Source, Fig. 2a). This ellipsoid is centered at the substrate surface and its width is matched to the experimentally measured transverse width of the melt pool. The dimensions of the ellipsoidal heat sources were determined from image-based measurements of metallographic cross-sections of the deposited bead. The cross-sectional images were processed to identify the fusion boundary and to extract the characteristic geometric parameters of the melt pool. The measured transverse width was used to define the dimensions of the upper ellipsoid, while the penetration depth and finger-like profile observed in the cross-section were used to calibrate the lower ellipsoid. This image-based procedure ensured that the heat source geometry remained consistent with the experimentally observed melt pool shape.

However, accurately simulating the characteristic finger-like penetration of the melt pool poses a greater challenge, especially in the absence of fluid flow modelling [30]. To overcome this, a second double-ellipsoidal heat source is incorporated, also following the Goldak formulation (Bottom Heat Source, Fig. 2a). The total process power is distributed between both ellipsoidal sources in proportion to their respective volumes, ensuring that the nominal input energy remains consistent with experimental conditions.

The analytical expressions corresponding to the thermal model proposed by Goldak for the front and rear sections of the double-ellipsoidal heat source are presented in Eqs. (1) and (2).

$$q_{vf}(x,y,z) = \frac{6\sqrt{3}f_f Q}{a_f bc\pi\sqrt{\pi}} e^{-3\frac{z^2}{c^2}} \cdot e^{-3\frac{y^2}{b^2}} \cdot e^{-3\frac{x^2}{a_f^2}} \quad (1)$$

$$q_{vr}(x,y,z) = \frac{6\sqrt{3}f_r Q}{a_r bc\pi\sqrt{\pi}} e^{-3\frac{z^2}{c^2}} \cdot e^{-3\frac{y^2}{b^2}} \cdot e^{-3\frac{x^2}{a_r^2}} \quad (2)$$

In the present study, these equations are applied independently to both the upper (surface) and lower (penetration) ellipsoidal sources described previously. The geometric parameters (Fig. 2) used in each ellipsoid (such as width (b), height (c), and rear and forward length (a_r and a_f, respectively)) were extracted from transverse cross-sections of the deposited bead and from top-view images of the final weld crater [18, 21]. The experimental measurements of the weld bead were obtained using a Python-based image processing program, enabling higher accuracy in extracting the melt pool and final crater dimensions. The potential parameter is represented in the equation with letter Q. An iterative calibration process was performed to refine the dimensions (particularly the width and depth) of the ellipsoids, with the objective of closely reproducing the experimentally observed finger morphology. The longitudinal dimensions used in the Goldak equations were determined based on the geometry of the final crater (Fig. 2b), following the methodology established in previous studies [18].

In the Goldak formulation, the front and rear fractions of heat distribution (f_f and f_r, respectively) define the asymmetry of the ellipsoidal heat input along the direction of deposition. In this study, the values $f_f = 1.4$ and $f_r = 0.6$ were adopted, consistent with literature and experimental observations. These values reflect the higher concentration of energy at the leading edge of the arc and the relatively smoother decay toward the rear, better capturing the directional heat flow and resulting melt pool shape typical of WAAM processes [31, 32].

Ellipsoid dimensions To accurately capture the geometry of elliptical features within the image data, a semi-automated method was implemented using Python, OpenCV, and Matplotlib. The procedure enables the user to manually select points along the boundary of a visible ellipse in an image and then fits a geometrically consistent ellipse to those points using a least-squares approach.

The user interactively selects points along the edge of a visually identified ellipse. The first four selected points are interpreted (for labeling and reference) as approximate left, right, top, and bottom extremes of the ellipse. Additional points improve the geometric accuracy of the fitted ellipse. Once a minimum of five points is selected, the user completes the selection by right-clicking, which triggers the ellipse fitting process (Fig. 3).

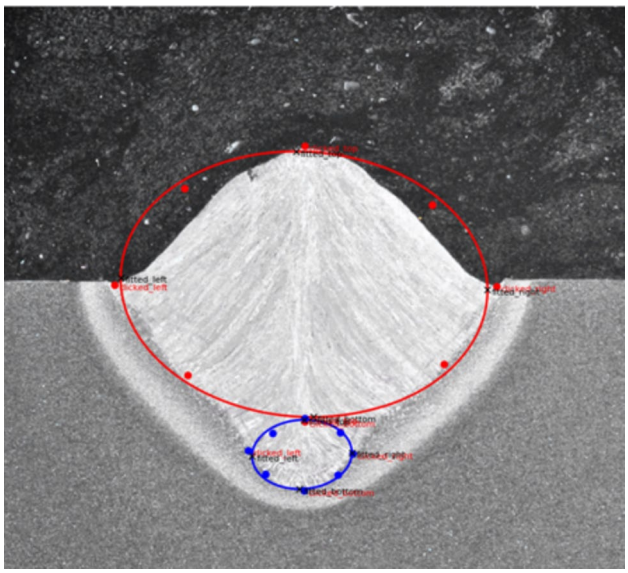


Fig. 3 Top and bottom ellipsoid measurement by image treatment in the experimentally deposited WAAM single bead on the substrate

The selected points are passed to OpenCV's fit Ellipse function, which fits a rotated ellipse using a least-squares minimization. The output is an ellipse defined by the following: center coordinates and lengths of the major and minor axes.

The resulting ellipse is rendered on the image using Matplotlib's Ellipse. The actual top, bottom, left, and right points of the fitted ellipse are computed from its parametric equation. These are determined by sampling the ellipse at high angular resolution (360 points) and identifying the maximum and minimum values in the x and y directions.

To enable pixel-to-millimeter conversion for geometric measurements, an automated tool to detect scale bars labeled in the image (1 mm) is developed. The method uses OCR combined with image processing to extract the physical length in mm and its corresponding length in pixels.

The image is preprocessed using grayscale conversion, contrast equalization, and bilateral filtering to enhance text clarity. Tesseract OCR is then applied to extract all visible text. A dedicated parser searches for digit labels that appear adjacent to the unit "mm", based on spatial proximity.

The conversion factor is then calculated as Eq. (3).

$$Scale\ Factor = \frac{n}{pixels} \quad (3)$$

where n (mm) is a numerical value (e.g., 1, 2, 3…) indicating the physical length of the scale bar in millimeters,

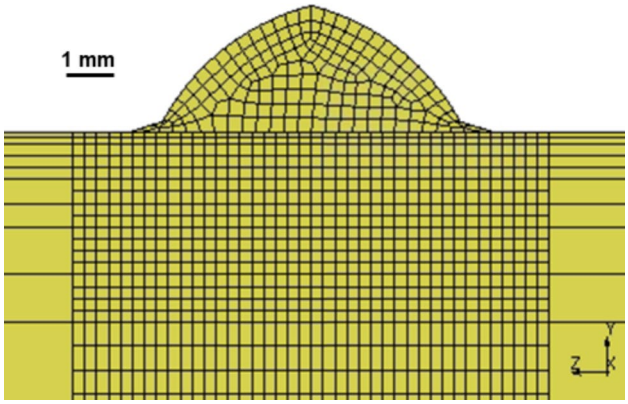


Fig. 4 Cross-sectional view of the weld bead and substrate geometry with the applied mesh in the FEM model

detected via OCR from the image and pixels is the measured length of the scale bar in the image, computed in pixels by analyzing the dark bar's start and end positions.

2.2.2 FEM model

The thermal simulation using FEM was conducted with the commercial software MSC Marc Mentat, known for its robust capabilities in nonlinear thermomechanical analysis. A three-dimensional mesh composed of hexahedral elements with eight nodes each was generated, with the full geometry consisting of approximately 181,000 elements. To ensure accurate temperature prediction (particularly in critical regions such as the melt pool and the HAZ) a locally refined mesh was applied in these areas. This refinement improves the resolution of steep thermal gradients, which are essential for capturing the rapid temperature changes that occur during deposition [33]. The mesh configuration adopted in this study is based on a previously mesh reported by the authors, ensuring numerical reliability while maintaining computational efficiency [18].

Moreover, the weld bead geometry was adapted to closely match the experimentally observed morphology (Fig. 4). This geometric fidelity allows the simulation to better replicate the thermal behavior of the actual process. In addition, at the interface between the bead and the substrate, a "Glue" contact condition was defined. This boundary condition ensures accurate heat transfer modeling by assuming perfect thermal contact without mechanical separation or sliding (an appropriate assumption for single-pass WAAM processes).

The weld path followed by the moving heat source was also defined in MSC Marc Mentat, replicating the deposition trajectory used experimentally. This synchronization ensures consistent energy input and spatial heat evolution. A total simulation time of 3.2 s was considered, with the time step in the program set to 0.01 s.

For the boundary conditions, convective and radiative heat losses were applied to all outer surfaces of the substrate and deposited bead exposed to the environment. These losses simulate realistic thermal exchanges during the WAAM process. Additionally, the model accounted for the deposition rate and wire feed speed during the process, ensuring the thermal load introduced by material addition was properly considered.

2.2.3 FDM model

In the FDM thermal model, the simulation domain was discretized into a structured Cartesian grid. The number of nodes was set to 121, 122, and 66 along the longitudinal (X), transverse (Y), and vertical (Z) directions, respectively. The spatial resolution in each direction (dx, dy, dz) was defined by dividing the total domain length by the number of intervals, ensuring uniform mesh spacing throughout the domain. The computational domain of the FDM model was discretized using a simple and uniform grid in order to reduce computational cost while preserving the reliability of the thermal predictions. Previous modelling studies indicated that this discretization strategy is sufficient for the present application. In the adopted mesh, the cell size was 1 mm in the x-direction and 0.5 mm in the y- and z-directions. This spatial resolution provided sufficiently reliable temperature predictions while considerably reducing the computational demand of the model [11]. The weld bead geometry was simplified and assumed to be semicircular in cross-section. Unlike the FEM approach, which closely follows the experimental morphology, this assumption also reduces the geometric complexity and contributes to lowering the computational cost while still capturing the essential thermal behavior.

After discretizing the geometry, logical masks were defined within the MATLAB-based FDM model to distinguish the different regions of the simulation domain (Fig. 5). Specifically, separate masks were generated for the substrate, the deposited bead, and the surface nodes of the substrate. The use of masks is essential in FDM modelling, as it allows the assignment of distinct material properties (e.g., thermal conductivity, density, specific heat) and boundary conditions to each region. Differentiating these domains is particularly important for accurately modelling the thermal behavior, since the heat transfer characteristics vary significantly between the substrate and the deposited material. Additionally, isolating the surface nodes enables precise application of heat flux from the heat source and more accurate calculation of convective and radiative losses at the substrate-air interface. To simplify the model, the outer surface of the

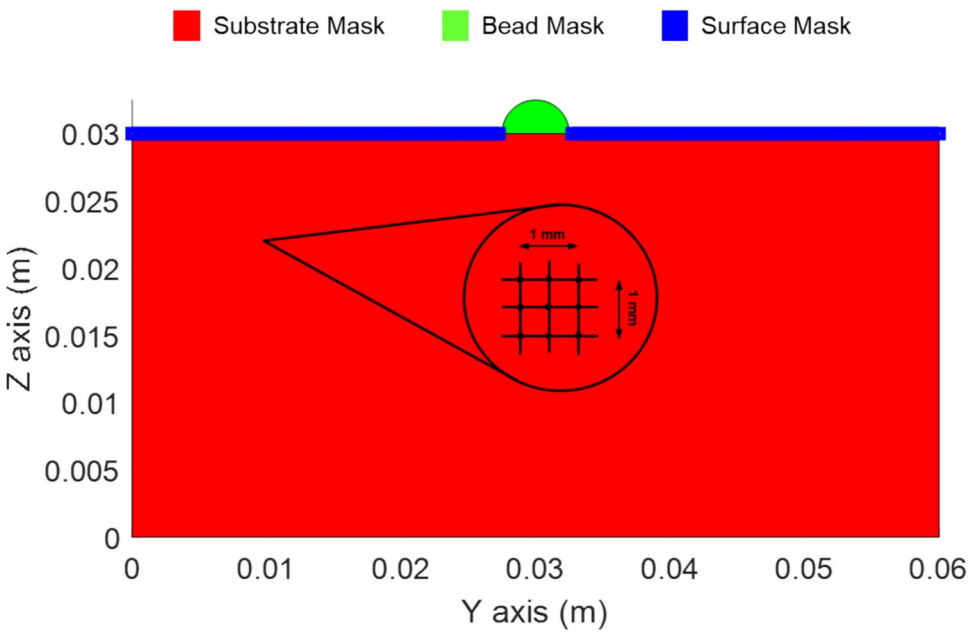


Fig. 5 Schematic representation of the substrate and weld bead geometry with the defined masks and discretization size used in the FDM model

bead was not considered as an exposed surface and thus was not subjected to convection or radiation boundary conditions. This simplification has minimal impact on the substrate region analyzed in this study.

The temporal discretization in the FDM model was defined by setting a total simulation time of 3.2 s, representing the full duration of the WAAM deposition process under study. A time step size of 0.0025 s was chosen to ensure numerical stability and capture the transient thermal behavior with sufficient temporal resolution [11]. This results in a total of 1280 discrete time steps for the simulation. To analyze the temperature distribution evolution, an intermediate time step was selected at approximately half of the total simulation duration, which was used for detailed visualization and comparison with experimental data. This careful choice of temporal parameters balances the need for accuracy with computational efficiency.

Taking into account the total simulation time, the initial position of the process on the substrate, and the specified travel speed of the torch, the motion of both heat sources was defined within the model. By providing the travel speed as an input parameter, the thermal sources move progressively along the longitudinal axis of the substrate, accurately replicating the real WAAM deposition path. This implementation ensures that the dynamic heat input is correctly distributed over time and space.

The simulation calculates the temperature evolution node by node using a structured loop system composed of three nested loops, each corresponding to one of the spatial dimensions (X, Y, and Z). At each iteration, the thermal state of the node is updated based on its material properties and the specific boundary conditions associated with its location (whether it belongs to the substrate, the bead, or the free surface). This localized treatment allows the model to apply the appropriate heat transfer equations (conduction, convection, and radiation where applicable).

Equation (4) defines the general heat transfer equation used throughout the simulation domain, expressed as the transient three-dimensional heat conduction equation.

$$\rho \cdot c_p \cdot \frac{\partial T}{\partial t} = \nabla \cdot (k \cdot \nabla T) + Q(x, y, z, t) \tag{4}$$

where ρ is the material density, c_p is the specific heat capacity, T is the temperature, k is the thermal conductivity, and Q is the internal heat generation rate from the moving heat source, which vary spatially and temporally based on the torch movement.

This approach ensures that the thermal load is applied dynamically, following the travel speed and positioning of the heat source throughout the simulation domain. This expression is discretized using finite differences in both time and space, enabling the node-by-node temperature evolution throughout the simulation. As in the FEM model, the thermal properties of the material are dependent on temperature.

To illustrate the thermal calculation procedure, the case of a node located on the substrate surface (i, j, k) is described in detail (Fig. 6). This figure represents a schematic illustration of the heat transfer mechanism at a node (i, j, k) in the surface of the substrate. This type of node is particularly relevant due to its exposure to both convective and radiative heat losses, in addition to conduction with neighboring nodes. Once this surface case is understood, the same logic can be extended to the rest of the domain,

adjusting the boundary conditions depending on whether the node lies within the substrate bulk, the deposited bead, or at internal interfaces.

At each time step n + 1, the finite-difference equation at a surface node (i, j, k) of the substrate can be formulated as follows:

$$\rho \cdot c_p \cdot \frac{{T_{i,j,k}}^{n+1} - {T_{i,j,k}}^{n}}{dt} \cdot dx \cdot dy \cdot dz = q_{cond1} \cdot dx \cdot dz + q_{cond2} \cdot dy \cdot dz + q_{cond3} \cdot dx \cdot dz + q_{cond4} \cdot dy \cdot dz + q_{cond5} \cdot dx \cdot dy - q_{conv} \cdot dx \cdot dy - q_{rad} \cdot dx \cdot dy + Q(x, y, z, t) \cdot dx \cdot dy \cdot dz \quad (5)$$

where q_{cond} represents the heat flux due to conduction, q_{conv} refers to convection heat flux and q_{rad} refers to the radiation heat flux.

The heat conduction in the discretized model is computed using a finite difference scheme. For a lateral face in the Y-direction, the conductive heat flux between adjacent nodes (i, j, k) and (i, j-1, k) is given by Fourier's law:

$$q_{cond1} = k_{i,j,k} \cdot \frac{{T_{i,j-1,k}}^{n} - {T_{i,j,k}}^{n}}{dy} \quad (6)$$

Similarly, for the bottom face in the Z-direction the conductive heat flux is defined as follows:

$$q_{cond5} = k_{i,j,k} \cdot \frac{{T_{i,j,k-1}}^{n} - {T_{i,j,k}}^{n}}{dz} \quad (7)$$

In addition to conductive heat transfer, convective and radiative heat losses to the environment are considered for surface nodes exposed to the inert atmosphere. These boundary conditions are implemented in the model through additional source terms.

The convective heat flux is defined by Newton's law of cooling:

$$q_{conv} = h \cdot ({T_{i,j,k}}^{n} - T_{\infty}) \quad (8)$$

where h is the convection coefficient and T_{∞} refers to the ambient temperature.

The radiative heat flux is expressed using the Stefan-Boltzmann law:

$$q_{rad} = \varepsilon \cdot \sigma \cdot \left[\left({T_{i,j,k}}^{n} \right)^4 - \left(T_{\infty} \right)^4 \right] \quad (9)$$

where ε is the emissivity of the material and σ is the Stefan-Boltzmann constant.

Thus, substituting these equations in Eq. (5), and dividing by dx·dy·dz, the following expression can be obtained:

$$\rho \cdot c_p \cdot \frac{{T_{i,j,k}}^{n+1} - {T_{i,j,k}}^{n}}{dt} = k_{i,j,k} \cdot \left(\frac{{T_{i-1,j,k}}^{n} - 2 \cdot {T_{i,j,k}}^{n} + {T_{i+1,j,k}}^{n}}{dx^2} + \frac{{T_{i,j-1,k}}^{n} - 2 \cdot {T_{i,j,k}}^{n} + {T_{i,j+1,k}}^{n}}{dy^2} + + \frac{{T_{i,j,k-1}}^{n} - {T_{i,j,k}}^{n}}{dz^2} \right) - \frac{h}{dz} \cdot ({T_{i,j,k}}^{n} - T_{\infty}) - \frac{\varepsilon \cdot \sigma}{dz} \left[\left({T_{i,j,k}}^{n} \right)^4 - \left(T_{\infty} \right)^4 \right] + Q(x, y, z, t) \quad (10)$$

This formulation allows temperature to be updated at each node and time step, capturing the transient thermal behavior of the WAAM process with surface heat losses and volumetric heating taken into account.

2.2.4 Computational environment

The thermal simulations for both the FEM and FDM models were performed on a workstation equipped with an Intel® Core™ i5-7500 CPU at 3.40 GHz (4 cores), 12 GB of RAM running at 2133 MHz, and Windows 11 Education as the operating system. The finite element simulations were developed using MSC Marc Mentat 2021.4 (Hexagon), while the finite difference model was implemented in MATLAB R2020a. A total process time of 3.2 s was simulated in both models. These specifications provide a consistent basis for evaluating and comparing the computational performance and efficiency of the two thermal modelling approaches.

2.3 Dimension measurements

To assess the predictive accuracy of the thermal models (FEM and FDM) in replicating the experimental melt pool geometry, a comprehensive dimensional study was conducted. This evaluation was centered on two fundamental morphological features: melt pool width and depth, derived from transverse metallographic sections of the single-pass deposition.

Adapting the strategy outlined by Fernández-Zabalza et al. [34], where they targeted the overall bead geometry, this work focuses on the melt pool region. Both the FEM and FDM simulation results were post-processed to delineate the

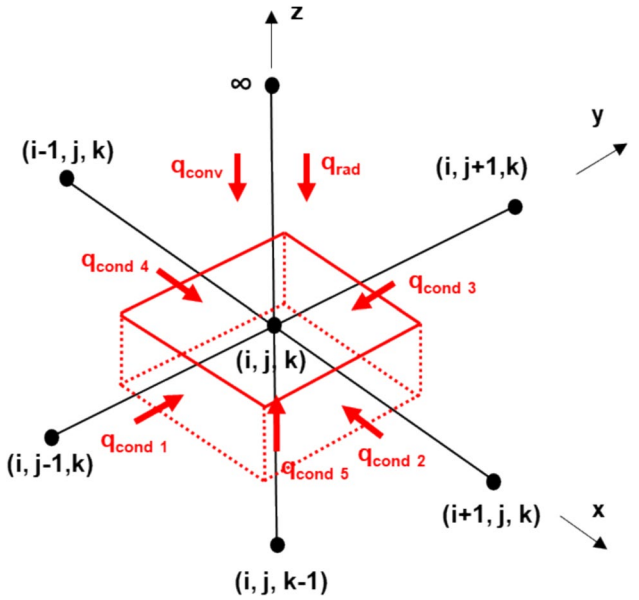


Fig. 6 Heat transfer mechanisms at a surface node in the discretized FDM domain

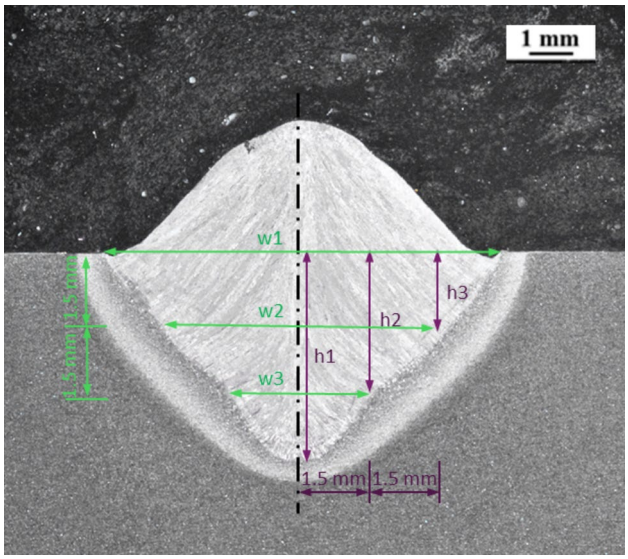


Fig. 7 Cross-sectional view of the experimentally deposited WAAM single bead on the substrate. The melt pool width and height dimensions used for comparison with the thermal models (FEM and FDM) are indicated

melt pool contours, based on thermal thresholds representative of the molten region at 1450 °C [25]. These boundaries provide insight into the extent of the melt pool. Three reference points were chosen across the width (w1, w2, and w3) and height (h1, h2, and h3) of the cross-section as can be seen in Fig. 7. This spatial sampling approach accounts for potential asymmetries or irregularities in the melt pool profile, enabling a more representative characterization of its geometry. Such a method mitigates the impact of local anomalies and enhances the robustness of the comparison between simulation and experiment [34].

Additionally, cross-sectional areas of both the melt pool and HAZ were measured to enable a more comprehensive comparison between experimental results and thermal predictions from both FEM and FDM models. These areas were determined using thermal thresholds representative of key metallurgical transformations that were defined by the phase transformation of the substrate, situated at ~ 900 °C [25].

Dimensional data were extracted through a Python-based image processing algorithm, which ensured precise and repeatable measurements of the cross-sectional features. This digital method reduces operator subjectivity and strengthens the reliability of the morphological analysis. The results of the simulation were then quantitatively compared against the experimental data across the defined locations. This allowed an accurate validation of each model's capability to reproduce the thermal gradients and energy distribution typical of the WAAM process.

3 Results

The following section presents the results obtained from the application of the developed thermal models (FEM and FDM) and their comparison with experimental data. First, a qualitative and quantitative analysis of the morphology of the melt pool and the HAZ is conducted, using experimental measurements as a reference. Key parameters such as width, height, and area of these regions are compared with the outcomes of both models to assess their accuracy. Subsequently, a study of the computational cost of each modelling approach is carried out, analyzing simulation time and resource efficiency.

3.1 Thermal model comparison

The qualitative comparison between the thermal models and the experimental results reveals notable insights into the accuracy of each simulation. In the case of the FEM model, the predicted melt pool morphology closely resembles the experimental observations, clearly distinguishing both the horizontal ellipsoidal region and the finger extension, as shown in Fig. 8a. This strong correlation highlights the effectiveness of the double ellipsoid heat source in replicating the physical characteristics of the deposition process. Regarding the morphology of the heat-affected zone, defined by the 900 °C isotherm, the FEM model also produces a similar profile; however, a slight curvature is observed in

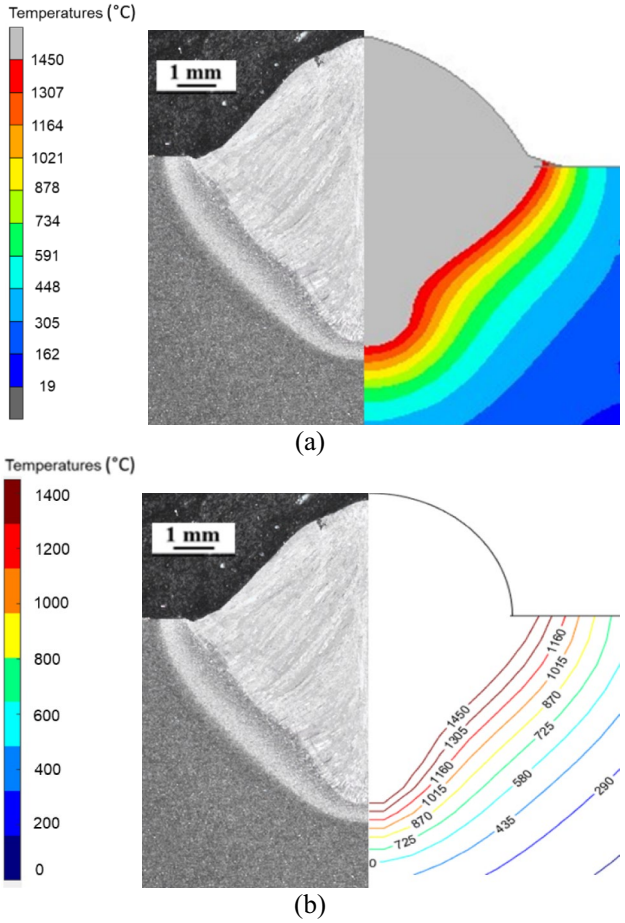


Fig. 8 Qualitative comparison of results between experiment and **a** FEM model; **b** FDM model

the mid-height region, which does not appear in the experimental cross-section. This discrepancy may be attributed to idealized boundary conditions, mesh-related interpolation effects, or simplifications in material property gradients in the numerical setup.

In contrast, the FDM model also captures the general shape of the melt pool with acceptable fidelity, differentiating the two heat source contributions, with less clarity than the FEM model (Fig. 8b). Interestingly, the HAZ morphology predicted by the FDM approach aligns more closely with the experimental result, particularly in the intermediate height region, where no pronounced curvature is present. This may be due to the non-realistic material properties and the structured mesh in the FDM model, which potentially leads to a more consistent heat distribution across the domain.

The graph shown in Fig. 9a presents a comparison of the melt pool and HAZ areas obtained experimentally and predicted by both thermal models. The melt pool areas calculated by FEM and FDM closely match the experimental measurements, demonstrating good predictive capability in this region. However, slight discrepancies are observed in the HAZ areas. The HAZ predicted by the FEM model shows a slightly smaller area compared to the experimental one, primarily due to the curvature observed in the mid-height region of the HAZ, which reduces the overall cross-sectional area. This curvature likely stems from the shape of the temperature distribution associated with the horizontal ellipsoid and local thermal gradients. On the other hand, the FDM model predicts a more vertically extended HAZ, slightly overestimating the total area. This overestimation may result from the simplified representation of the heat source and the more homogeneous propagation of heat in the structured mesh, particularly in the vertical direction [21].

Figure 9b shows the comparison of melt pool width at three different locations (w1, w2, and w3). The FDM model demonstrates good agreement with the experimental values across all three width parameters. In the case of the FEM model, small deviations are observed in w1 and w2, corresponding to the region closer to the substrate surface, which is predominantly influenced by the horizontal ellipsoid. These deviations can be attributed to the sensitivity of the FEM simulation to mesh resolution and heat source parameterization, especially in the near-surface region where high thermal gradients and rapid cooling occur. Accurately capturing the lateral extent of the melt pool in this region may require even finer meshing or more refined calibration of the ellipsoid parameters [21, 35].

Finally, Fig. 9c compares the melt pool height at three different vertical positions (h1, h2, and h3). Both FEM and FDM models show excellent agreement with the experimental measurements in all three cases, with negligible deviations. This indicates that the vertical heat penetration and thermal diffusion have been effectively captured by both modelling approaches for the melt pool.

The good agreement between the numerical results and experimental melt pool geometries can be explained by the suitability of the double ellipsoidal heat source to represent the asymmetric nature of arc-based energy distribution in WAAM. The model proposed by Goldak accounts for the non-uniform heat input between the front and rear regions of the molten pool, allowing for a realistic representation of both the wider surface melting and the deeper penetration zone. In the present work, this capability is further enhanced by the use of a double double-ellipsoidal configuration, also employed by Azar et al. and Kik et al., which enables independent adjustment of surface spreading and finger-like penetration effects observed experimentally [19–21]. By calibrating the geometric parameters using image-based measurements of the melt pool boundaries, the heat source closely reproduces the actual thermal footprint of the process, resulting in accurate predictions of both melt pool shape and heat-affected zone dimensions.

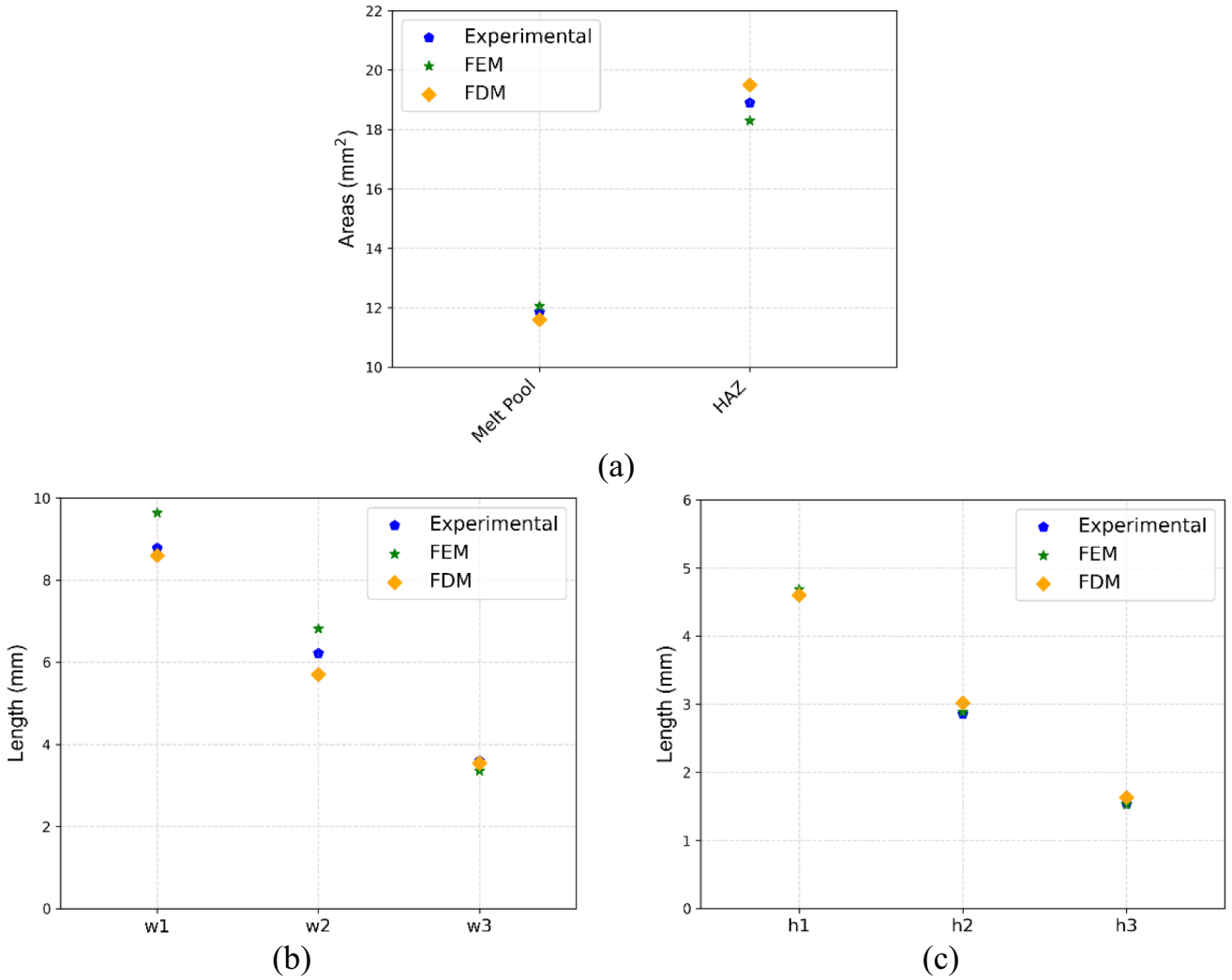


Fig. 9 Quantitative comparison of results between experiment and FEM and FDM models; **a** melt pool and HAZ areas; **b** melt pool width; **c** melt pool height

Table 1 Comparison of computational performance metrics between FEM and FDM thermal models

Metric	FEM	FDM	Notes
Simulation time (s)	19,193	612	FDM is significantly faster, reducing total simulation time by over 30×
Peak memory (GB)	10.5	8.9	FDM uses less memory, making it more efficient for lower-spec hardware
Time step size (s)	0.01	0.0025	FDM uses smaller time steps, requiring more iterations but remaining faster overall
Total time steps	320	1280	Higher time step count in FDM due to finer temporal resolution, but not a performance issue
CPU cores used	4	4	Same number of cores used, fair baseline for computational comparison
Preprocessing time (s)	129	7	FDM benefits from minimal setup time due to its simplified domain and direct coding approach

3.2 Computational cost evaluation

To thoroughly assess the computational efficiency of the two thermal modelling approaches (FEM and FDM) a comparison was conducted based on several performance metrics (Table 1). Both models simulated a total process time of 3.2 s under identical hardware conditions. The FEM model required a total simulation time of 19,193 s (approximately 5.3 h), whereas the FDM model completed the simulation in only 612 s (just over 10 min), representing a drastic

reduction in computational time. This significant difference is attributed to the simpler numerical scheme of FDM and the more explicit structure of the algorithm, which avoids some of the computational overhead associated with FEM (matrix assembly and global system solving).

In terms of memory usage, FEM exhibited a slightly higher peak memory demand at 10.5 GB, compared to 8.9 GB for FDM. This is likely due to the denser data structures and more complex mesh management inherent in FEM simulations, especially when using 3D hexahedral elements with refined meshing near the melt pool and HAZ zones. Both simulations were executed using 4 CPU cores, ensuring a fair and consistent comparison.

The time step size used in each model also reflects differences in numerical stability and convergence. FEM operated with a time step of 0.01 s, resulting in a total of 320 time steps, while FDM employed a smaller time step of 0.0025 s, leading to 1280 time steps. Despite the higher number of time steps in FDM, the overall simulation remained considerably faster, further highlighting the computational simplicity and efficiency of the finite difference scheme.

Notably, preprocessing time (such as mesh generation, model setup, and boundary condition definition) was substantially longer in the FEM case (129 s) compared to FDM (7 s), largely due to the complexity of defining a 3D mesh and path-dependent heat input in commercial FEM software.

The significantly lower computational cost of the FDM model compared to FEM is primarily attributed to its simpler spatial discretization and reduced numerical complexity. Unlike FEM, which requires mesh generation, shape functions, and the solution of large sparse matrix systems, the FDM approach operates on a structured and uniform grid with explicit or simplified finite difference formulations. This reduces preprocessing effort and computational overhead per time step. Despite this simplification, the model remains accurate for WAAM thermal simulations because heat transfer in the solid region is predominantly governed by diffusion, which is well captured by structured finite difference schemes when an appropriate spatial resolution is used [11]. Additionally, since steep gradients are mainly confined to the melt pool and near-field region, the selected grid resolution is sufficient to resolve the dominant thermal behavior without the need for the higher geometric flexibility offered by FEM.

Overall, the comparison clearly demonstrates that the FDM model offers significant advantages in terms of computational speed and resource efficiency. These attributes make it particularly suitable for industrial applications where rapid simulation feedback is essential, such as process parameter optimization, real-time control strategies, or iterative design loops. While FEM provides greater geometric flexibility and higher fidelity near interfaces, the trade-off in computational demand can be prohibitive in high-throughput environments, reinforcing the practicality of FDM as a valuable alternative.

4 Conclusions

Based on the experimental and numerical findings, the following conclusions can be established:

- Both thermal models successfully replicate the melt pool geometry observed experimentally. The FEM model provides a more detailed representation, clearly distinguishing the horizontal ellipsoid and finger structure. The FDM model also captures this morphology but in a more simplified form due to the geometric assumptions made.
- Regarding the heat-affected zone, the FEM model slightly underestimates the area due to a mid-height curvature not observed experimentally, while the FDM model slightly overestimates the area because of a greater predicted height.
- From a dimensional standpoint, the melt pool widths and heights obtained from both models align well with the experimental measurements.
- The comparative analysis of computational performance revealed a significant difference in efficiency between the two thermal modelling approaches. While both FEM and FDM achieved high correlation with the experimental results, the FDM model demonstrated a markedly lower computational cost, reducing total simulation time by over 30×.

Author contributions Xabier Sandua contributed to conceptualization, methodology, software development, validation, investigation, and writing of the original draft. Juan Sustacha contributed to software development, formal analysis, investigation, and writing of the original draft. Miguel Arizmendi contributed to conceptualization, methodology, and writing through review and editing. Fernando Veiga contributed to conceptualization, methodology, writing through review and editing, and funding acquisition. Alfredo Suárez contributed to conceptualization and investigation.

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Data availability Not applicable.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

1. Chernyshikhin SV et al (2023) In-plane measurements and computational fluid dynamics prediction of permeability for biocompatible NiTi gyroid scaffolds fabricated via laser powder bed fusion. IJB 10(1):0119. https://doi.org/10.36922/ijb.0119
2. Huang Y et al (2026) Formation mechanisms on amorphous-enhanced interfaces of Al/steel bimetallic freeform components via wire-based friction stir additive manufacturing. Addit Manuf 119:105113. https://doi.org/10.1016/j.addma.2026.105113
3. Peng J et al (2025) Effect of the activator B(OCH3)3 on the microstructure and mechanical properties of Cu-Mn-Al alloy coating via CMT cladding. Crystals 15(10):881. https://doi.org/10.3390/cryst15100881
4. Aldalur E, Veiga F, Suárez A, Bilbao J, Lamikiz A (2020) High deposition wire arc additive manufacturing of mild steel: strategies and heat input effect on microstructure and mechanical properties. J Manuf Process 58:615–626. https://doi.org/10.1016/j.jmapro.2020.08.060
5. Evans SI, Wang J, Qin J, He Y, Shepherd P, Ding J (2022) A review of WAAM for steel construction – manufacturing, material and geometric properties, design, and future directions. Structures 44:1506–1522. https://doi.org/10.1016/j.istruc.2022.08.084
6. Lian R, Xu X, Zhao K, Fei X, Cao W, Zhu J (2026) Advancements in laser processes for additive manufacturing: techniques and future directions. J Mater Res Technol 41:2428–2451. https://doi.org/10.1016/j.jmrt.2026.01.173
7. Derekar K, Lawrence J, Melton G, Addison A, Zhang X, Xu L (2019) Influence of interpass temperature on wire arc additive manufacturing (WAAM) of aluminium alloy components. MATEC Web Conf 269:05001. https://doi.org/10.1051/matecconf/201926905001
8. Gurmesa FD, Lemu HG (2023) Literature review on thermomechanical modelling and analysis of residual stress effects in wire arc additive manufacturing. Metals 13(3):526. https://doi.org/10.3390/met13030526
9. Ahmad SN, Manurung YH, Mat MF, Minggu Z, Leitner M (2020) FEM simulation procedure for distortion and residual stress analysis of wire arc additive manufacturing. IOP Conf Ser Mater Sci Eng 834(1):012083. https://doi.org/10.1088/1757-899X/834/1/012083
10. Mahmood MA, Tariq U, Oane M, Liou F (2024) Analytical and FEM models for thermal analysis and residual stresses using wire arc-based welding and additive manufacturing of SUS304. Int J Adv Manuf Technol 133(5–6):2363–2380. https://doi.org/10.1007/s00170-024-13797-0
11. Stockman T, Schneider JA, Walker B, Carpenter JS (2019) A 3D finite difference thermal model tailored for additive manufacturing. JOM 71(3):1117–1126. https://doi.org/10.1007/s11837-019-03338-6
12. Singh V, Kunal G, Rooprai RS, Bansal A, Vishnoi M, Singla AK (Sep.2024) FEA-guided development of Inconel 625 clad on mild steel using WAAM and their cavitation erosion resistance. Int J Interact Des Manuf. https://doi.org/10.1007/s12008-024-02099-0
13. Yang Y, Van Keulen F, Ayas C (2020) A computationally efficient thermal model for selective laser melting. Addit Manuf 31:100955. https://doi.org/10.1016/j.addma.2019.100955
14. Prajadhiama KP, Manurung YH, Minggu Z, Pengadau FH (2019) Development of bead modelling for distortion analysis induced by wire arc additive manufacturing using FEM and experiment. MATEC Web Conf 269:05003. https://doi.org/10.1051/matecconf/201926905003
15. Cichota R (2004) Testing a finite-difference model for soil heat transfer by comparing numerical and analytical solutions. Environ Model Softw 19(5):495–506. https://doi.org/10.1016/s1364-8152(03)00164-6
16. Grzesik W, Bartoszuk M, Nieslony P (2004) Finite difference analysis of the thermal behaviour of coated tools in orthogonal cutting of steels. Int J Mach Tools Manuf 44(14):1451–1462. https://doi.org/10.1016/j.ijmachtools.2004.05.008
17. Qiao J-H, Bolot R, Liao H, Bertrand P, Coddet C (2013) A 3D finite-difference model for the effective thermal conductivity of thermal barrier coatings produced by plasma spraying. Int J Therm Sci 65:120–126. https://doi.org/10.1016/j.ijthermalsci.2012.09.008
18. Sandua X et al (2025) Implementación de un modelo térmico para la estimación de la geometría de la zona afectada térmicamente en la fabricación aditiva por arco con hilo (WAAM) mediante análisis FEM e integración de redes neuronales artificiales (ANN). Revista DYNA 100(6):568–576. https://doi.org/10.52152/D11486
19. Kik T, Garašić I, Perić M, Landek D, Jurica M, Tonković Z (2024) Modifications of the heat source model in numerical analyses of the metal-cored arc welding process. Energy 302:131811. https://doi.org/10.1016/j.energy.2024.131811
20. Goldak J, Chakravarti A, Bibby M (1984) A new finite element model for welding heat sources. Metall Trans B 15(2):299–305. https://doi.org/10.1007/BF02667333
21. Azar AS, Ås SK, Akselsen OM (2012) Determination of welding heat source parameters from actual bead shape. Comput Mater Sci 54:176–182. https://doi.org/10.1016/j.commatsci.2011.10.025
22. Ck N, Jhavar S (2025) Substrate-centric comparative analysis and validation of finite element and empirical models for residual stress prediction in remanufactured WAAM-fabricated SS316 parts. Multiscale and Multidiscip. Model. Exp. and Des. 8(1):112. https://doi.org/10.1007/s41939-024-00690-2
23. Zargari HH, Khajeh Mohammadilar SA, Ziaei Laleh V (2025) A long-term and pragmatic numerical simulation of wire arc additively manufactured ER70S6 and ER316L thin-wall via finite difference method. Prog Addit Manuf 10(12):11647–11663. https://doi.org/10.1007/s40964-025-01307-x
24. Ding J, Colegrove P, Mehnen J, Williams S, Wang F, Almeida PS (2014) A computationally efficient finite element model of wire and arc additive manufacture. Int J Adv Manuf Technol 70(1–4):227–236. https://doi.org/10.1007/s00170-013-5261-x

25. Shamlooei M, Zanon G, Brugnolli M, Bursi OS (2023) Study of the effects of laser cutting processes on S235 structural steel. ce papers 6(3–4):2521–2526. https://doi.org/10.1002/cepa.2566
26. Zhai W, Guo Y, Aishwarya, Canturri C, Shandro R, Zhou W (2024) Wire arc additive manufacturing of ER70S-6/S355 bimetal component. Mater Sci Eng A 900:146498. https://doi.org/10.1016/j.msea.2024.146498
27. Hu X, Chiu LNS, Huang A, Liu M, Yan W (2024) Application of melt pool profiles for parameter calibration of Goldak's heat source model. Addit Manuf 92:104379. https://doi.org/10.1016/j.addma.2024.104379
28. Gupta DK, Mulik RS (2025) Numerical simulation and experimental investigation of temperature distribution during the wire arc additive manufacturing (WAAM) process. Prog Addit Manuf 10(1):631–645. https://doi.org/10.1007/s40964-024-00647-4
29. Atkins G, Thiessen D, Nissley N, Adonyi Y (2002) Welding process effects in weldability testing of steels. Weld J 81(4):61–s. https://www.researchgate.net/publication/291433447
30. Zhao W, Wei Y, Long J, Chen J, Liu R, Ou W (2021) Modeling and simulation of heat transfer, fluid flow and geometry morphology in GMAW-based wire arc additive manufacturing. Weld World 65(8):1571–1590. https://doi.org/10.1007/s40194-021-01123-1
31. Moslemi N, Gohery S, Abdi B, Sudin I, Ayob A, Rhee S (2022) A novel systematic numerical approach on determination of heat source parameters in welding process. J Mater Res Technol 18:4427–4444. https://doi.org/10.1016/j.jmrt.2022.04.039
32. Hwang S-Y, Lee J-H, Kim S-C (2012) Numerical simulation of welding residual stress distribution on T-joint fillet structure. Int J Ocean Syst Eng 2(2):82–91. https://doi.org/10.5574/IJOSE.2011.2.2.082
33. Ke WC, Oliveira JP, Cong BQ, Ao SS, Peng B, Zeng Z (2022) Multilayer deposition mechanism in ultra high-frequency pulsed wire arc additive manufacturing (WAAM) of NiTi shape memory alloys. Addit Manuf 50:102513. https://doi.org/10.1016/j.addma.2021.102513
34. Fernández-Zabalza A, Veiga F, Suárez A, Uralde V, Sandua X, Alfaro JR (2025) Application of symmetric neural networks for bead geometry determination in wire and arc additive manufacturing (WAAM). Symmetry Basel 17(3):326. https://doi.org/10.3390/sym17030326
35. Kik T (2020) Heat source models in numerical simulations of laser welding. Materials Basel 13(11):2653. https://doi.org/10.3390/ma13112653

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